Merlin Epzibah. S
(192521331)

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0720

81%

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I Merlin Epzibah *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses.**
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation.**
2. Expand the compressed address back into its **full hexadecimal representation.**
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin g of the scenario and context(2)	Shows good understandi ng with minor gaps(1.5)	Partial understandin g; misses key aspects(1)	Misinterpret s or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1 .5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1. 5)	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification (1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear,	Generally clear with	Somewhat unclear or	Disorganize d and

	(20%) Understanding Scenario	(20%) Identification	Appln (25%) concepts	(20%) Decision Making	Clarity (15%)	
Q1	1.5	1.5	2	2	1.5	8.5
Q2	2	2	1.5	1.5	1	8
Q3	1	1.5	1.5	2	1.5	7.5
Q4	1.5	1.5	2	2	1.5	8.5
Q5	2	2	1.5	1.5	1	8

40.5

structured, and logically presented(1.5)

minor issues(1)

poorly structured(0.5)

difficult to understand

Question 1 – Smart Campus Subnetting

1. Question / Scenario

The Smart Campus has 192.168.100.0/24. School of Computing uses /25, School of Electronics uses /26, and School of Mechanical Engineering uses /27. The required tasks are network/broadcast addresses, valid host ranges, usable hosts, and the unused address range.

2. Given Information

Department	CIDR	Subnet Mask
School of Computing	/25	255.255.255.128
School of Electronics	/26	255.255.255.192
School of Mechanical Engineering	/27	255.255.255.224

3. Concept / Formula Used

Usable hosts = 2^h − 2. A /25 has 7 host bits, a /26 has 6 host bits, and a /27 has 5 host bits. Allocate the given subnets sequentially from the start of the /24 block so they do not overlap.

4. Step-by-Step Solution

1. Computing /25: block size 128 → 192.168.100.0–192.168.100.127.
2. Electronics /26: next boundary .128, block size 64 → 192.168.100.128–192.168.100.191.
3. Mechanical /27: next boundary .192, block size 32 → 192.168.100.192–192.168.100.223.
4. Unused space starts at .224 and ends at .255.

5. Final Answer

Department	CIDR	Subnet Mask	Network	First Host	Last Host	Broadcast	Usable Hosts
Computing	/25	255.255.255.128	192.168.100.0	192.168.100.1	192.168.100.126	192.168.100.127	126
Electronics	/26	255.255.255.192	192.168.100.128	192.168.100.129	192.168.100.190	192.168.100.191	62
Mechanical	/27	255.255.255.224	192.168.100.192	192.168.100.193	192.168.10	192.168.100.223	30

					0.222		

Remaining unused range
192.168.100.224–192.168.100.255 (32 total addresses).

Smart Campus - IPv4 Subnet Allocation

192.168.100.0/24 = 256 total addresses

Usable hosts: 126 + 62 + 30

Unused range: 192.168.100.224–192.168.100.255 (32 addresses)

Figure 1: Smart Campus IPv4 subnet allocation.

6. Conclusion

The three subnets are non-overlapping and provide 126, 62 and 30 usable hosts. The final 32 addresses remain unused.

Question 2 – Multinational Software Company Subnetting

1. Question / Scenario

The company has 10.10.0.0/16 and has allocated /24 to Software Development, /25 to Quality Assurance, /26 to Human Resources, and /27 to Executive Management.

2. Given Information

Department	CIDR	Subnet Mask
Software Development	/24	255.255.255.0
Quality Assurance	/25	255.255.255.128
Human Resources	/26	255.255.255.192
Executive Management	/27	255.255.255.224

3. Concept / Formula Used

A /16 contains $2^{(32-16)} = 65{,}536$ total IPv4 addresses. The subnets are allocated sequentially using their natural boundaries.

4. Step-by-Step Solution

5. Software Development /24: 10.10.0.0–10.10.0.255.
6. Quality Assurance /25: 10.10.1.0–10.10.1.127.
7. Human Resources /26: 10.10.1.128–10.10.1.191.
8. Executive Management /27: 10.10.1.192–10.10.1.223.
9. Unused space: 10.10.1.224–10.10.255.255.

5. Final Answer

Department	CIDR	Subnet Mask	Network	First Host	Last Host	Broadcast	Usable Hosts
Software Development	/24	255.255.255.0	10.10.0.0	10.10.0.1	10.10.0.254	10.10.0.255	254
Quality Assurance	/25	255.255.255.128	10.10.1.0	10.10.1.1	10.10.1.126	10.10.1.127	126
Human Resources	/26	255.255.255.192	10.10.1.128	10.10.1.129	10.10.1.190	10.10.1.191	62
Executive Management	/27	255.255.255.224	10.10.1.192	10.10.1.193	10.10.1.222	10.10.1.223	30

Remaining unused range

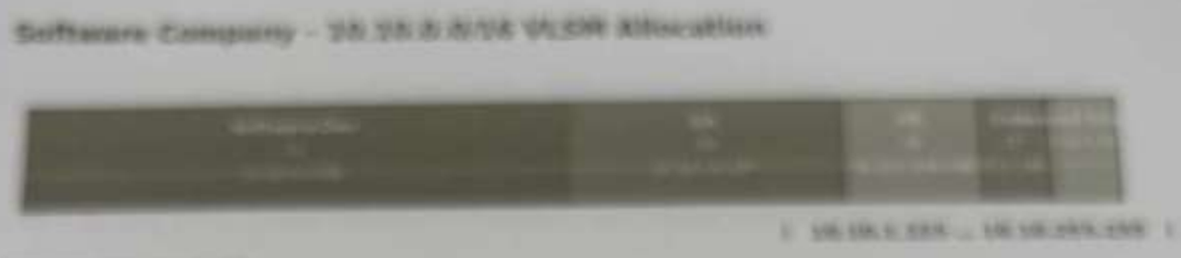

Figure 2: Software company IPv4 VLSM allocation

6. Conclusion

The assigned departments use 768 addresses in total, leaving 65,312 addresses available in the original /16 block.

Question 3 – Smart Hospital IPv6

1. Question / Scenario

The hospital is assigned 2001:0db8:abcd:0000:0000:0000:0000:0000/64 for its IPv6 network.

2. IPv6 Compression

Remove leading zeros in each hextet and compress the longest consecutive run of zero hextets.

Compressed notation
2001:db8:abcd::/64

3. Expanded Representation

Expanded notation
2001:0db8:abcd:0000:0000:0000:0000:0000/64

4. Network Prefix

The first 64 bits are the network prefix: 2001:0db8:abcd:0000. The remaining 64 bits are the interface identifier (host portion).

5. Host Address Calculation

10. IPv6 address length = 128 bits.
11. Host/interface bits = 128 − 64 = 64 bits.
12. Total possible addresses = 2^64 = 18,446,744,073,709,551,616.

Final answer

Network prefix = 64 bits; interface/host portion = 64 bits; total possible addresses in the /64 = 18,446,744,073,709,551,616.

IPv6 /64 Address Structure

2001:0db8:abcd:0000 : 0000:0000:0000:0000

Compressed: 2001:db8:abcd::/64

2^64 = 18,446,744,073,709,551,616 possible addresses